GGG-S-285b
May 22, 1964

SUPERSEDING
Int. Fed. Spec. GGG-S-00285a (GSA-FSS)
August 16, 1963

FEDERAL SPECIFICATION

SHEARS, PRUNING, HAND, (TWO-HAND OPERATED, LOPPING)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two-hand operated pruning or lopping shears capable of lopping low-hanging tree branches of 1, 1-1/4, 1-1/2, and 1-3/4 inches in diameter.

1.1.1 *Federal specification coverage.* Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 *Type and sizes.* The pruning shears covered by this specification shall be of the following sizes, as specified (see 6.1):

Size 20.—Overall length 20 inches minimum.
Size 24.—Overall length 24 inches minimum.
Size 27.—Overall length 27 inches minimum.
Size 30.—Overall length 30 inches minimum.

1.2.2 *Handle designs.* Unless otherwise specified (see 6.1), the shear handles shall be of any of the following designs:

Design 1.—Long metal ferruled wood handles (figure 1).
Design 2.—Solid metal handles with wood grips (figure 2).
Design 3.—Tubular metal handles (figure 3).
Design 4.—Bolted or riveted wood handles (figure 4).

1.2.3 *Cutting head styles.* Unless otherwise specified (see 6.1), the cutting heads shall be any of the following styles:

Style A.—Convex blade; and concave hook (figure 5).
Style B.—Straight blade and anvil (figure 6).

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specification, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specification required by activities outside

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the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-H-15424—Hand Tools, Packaging of.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

MIL-STD-130—Identification Marking of U. S. Military Property.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Wood handles and grips.* All wood handles and wood handgrips shall have a moisture content not exceeding 15 percent, shall have a slope of grain not exceeding 1 inch in 15 inches, and shall be free from defects except that slight stains, streaks, and discolorations will be allowed. The two handles or handgrips of any shears shall be of the same species of wood.

3.1.2 *Cutting blade.* The cutting blade shall be of high-grade carbon tool steel or alloy tool steel and shall show a hardness of not less than 48 nor more than 56 on the Rockwell C scale, the reading to be taken as near the cutting edge as practicable. The cutting blade shall have a ground and polished keen cutting edge.

3.1.2.1 *Bolt or rivet.* Unless otherwise specified (see 6.1), either a bolt or rivet shall be furnished for pivoting of the cutting head. The bolt shall be of high-grade steel properly heat-treated to withstand wear and deformation under maximum usage, and adequate means shall be provided for preventing loss of proper tightness. The rivet shall be of good quality iron or steel, shall fill the hole in both components, and shall be neatly set.

3.1.3 *Rubber.* When rubber is hereinafter specified, it shall be of natural, synthetic, or a compound of natural and synthetic rubber of a good commercial quality.

3.2 Finish.

3.2.1 *Wood handles and handgrips.* All wood handles and handgrips shall be smoothly finished so as to be comfortable to the hands. They shall be adequately covered with at least two coats of transparent varnish, shellac, lacquer, or similar material to prevent absorption of moisture and permit visible inspection of the wood. Provided all other requirements of this paragraph are adhered to, flame-treated or stained wood is acceptable.

3.2.2 *Solid metal handles.* The solid metal handles (design 2) shall be forged in accordance with good commercial practice, by employing drop forging equipment. All burrs and sharp edges shall be removed and all surfaces from the handgrip to within at least 3 inches of the cutter head shall be coated with paint or enamel, the color to be in accordance with the manufacturer's commercial practice.

3.2.3 *Tubular metal handles.* Tubular metal handles (design 3) shall be either of corrosion resisting metal or of metal made corrosion resisting by chrome or nickel plating.

3.3 General design and construction. The design of the shears shall be such that only normal manual effort is required by the operator to completely sever the capacity di-

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ameter branches specified. The shears shall be so constructed that no part will become detached while in use in normal service.

3.3.1 *Safety requirements.* Positive stop(s), shock absorbing stops, or other means shall be provided to prevent the handles from closing on the operator's hands when a branch is severed. When the shears are in a closed position, the cutting blade shall be in such a position that its point and cutting edge will be protected in a manner that the operator cannot injure his hands in the event that he picks up the shears by the cutting end.

3.3.2 *Optional features.* At the option of the manufacturer, commercially available features may be furnished on pruning shears such as the following:

(*a*) The inside on the hook may be serrated to prevent branch slippage.

(*b*) A draw-cut design may be furnished, which imparts a sliding movement to the blade.

(*c*) Compound leverage may be furnished at the cutting head, which reduces the manual effort exerted by the operator.

(*d*) Any other commercially available feature which has been used successfully in the open market.

3.3.3 *Ferrules and caps.* Ferrules and caps used on wood handles or handgrips shall be of such strength and so assembled as to act as a rigid support and prevent the handles from breaking, splitting, and loosening in service. The ferrules and caps shall be securely attached so as not to loosen in service.

3.4 Identification marking.

3.4.1 *Civil agencies.* All shears purchased for civil agencies shall be marked in a plain and permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

3.4.2 *Military agencies.* In addition to the marking specified in 3.4.1, all shears shall be marked in accordance with MIL-STD-130.

3.5 Illustrations. The illustrations shown herein are for the purpose of identification and are not intended to preclude the purchase of two-hand operated pruning or lopping shears which otherwise conform to the specification requirements. It is recommended that no options be exercised on the various handle designs and cutting head styles, so as to obtain a greater amount of bids

3.6 Handles and shanks. All handles and shanks shall be of such design and construction as to provide for cutting the capacity size branches specified without breaking, bending to the extent of permanent deformation, or in any other manner become defective.

3.6.1 *Handle design No. 1.* Handle design No. 1 (figure 1) shall have wood handles and long metal ferrules with or without caps (see 3.3.3). The wood handles shall be any of the following species: White ash, hickory, hard maple, pecan, oak, or birch. The metal ferrules shall be of the strap type, seamless type, or welded seam type. Pins or rivets used to secure the handle in the tang shall extend entirely through the wood handles and tang in a manner which shall prevent them from becoming loose.

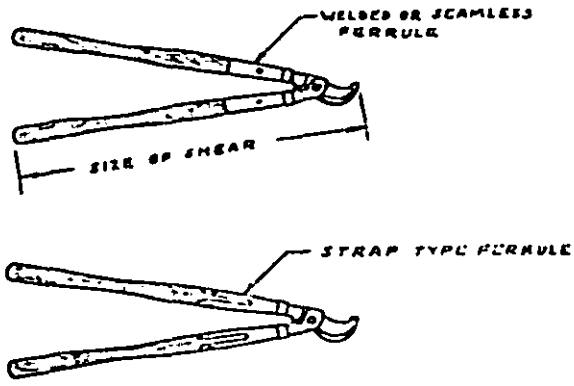


FIGURE 1.—Handle design No. 1
Long metal ferruled wood handles.

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3.6.2 *Handle design No. 2.* Handle design No. 2 (figure 2) shall have metal handles and wood handgrips with metal ferrules. The wood handgrips shall be any of the following species: white ash, hickory, hard maple, pecan, oak, or birch. The tangs of the shears shall either extend through the handgrips or extend into the handgrips or handles to such a depth as to preclude their loosening in service. The metal shanks shall be taper forged from a large section near the cutting head, down to a small section near the handgrip, so as to provide a well-balanced shear.

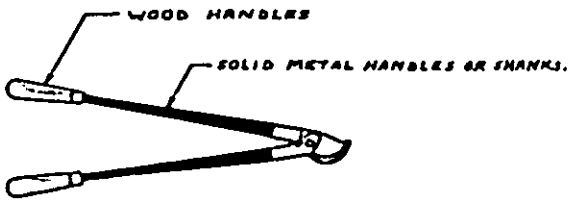


FIGURE 2.—Handle design No. 2
Solid metal handle with wood grips.

3.6.3 *Handle design No. 3.* Handle design No. 3 (figure 3) shall have tubular metal handles or shanks with either rubber or wood handgrips. The tubular handle or shank shall be securely attached to the cutting head tangs in a manner to prevent loosening in service. When furnished, rubber handgrips shall enclose the end of the tubular handle and shall tightly adhere to the handle. The overall length of the handgrips shall be not less than 4 inches. When furnished, wood handgrips or handles shall be of the following species: white ash, hickory, hard maple, pecan, oak, or birch.

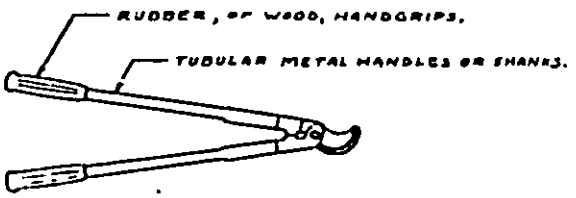


FIGURE 3.—Handle design No. 3
Tubular metal handles.

3.6.4 *Handle design No. 4.* Handle design No. 4 (figure 4) shall have wood handles securely fastened to the cutting head by employing at least two bolts or two rivets in each handle. The wood handles shall be either of white ash or hickory species.

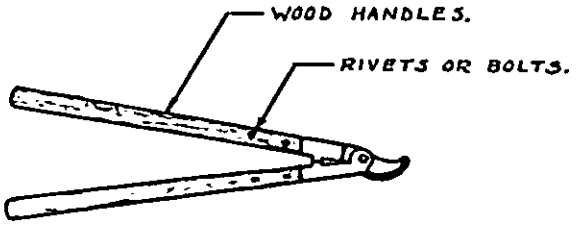


FIGURE 4.—Handle design No. 4
Bolted or riveted wood handles.

3.7 Cutting head.

3.7.1 *Style A.* Style A cutting head (figure 5) shall be furnished with a blade having a convex shaped cutting edge periphery, and a concave hook to cradle the tree branches during the pruning operation. The blade and hook shall be of forged steel. The cutting head shall be of the side-cut-action type.

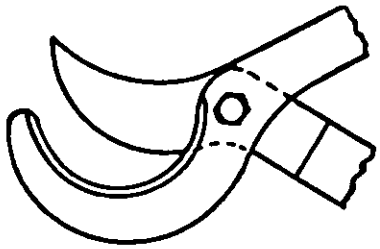


FIGURE 5.—Style A, convex blade; and concave hook cutting head, shown partially open.

3.7.2 *Style B.* Style B, straight blade and anvil cutting head (figure 6) shall be of the center-cut-action type. The blade and anvil shall be of the replaceable types. The anvil shall be made of brass or bronze.

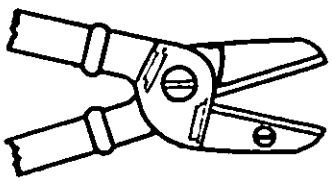


FIGURE 6.—Style B, straight blade and anvil cutting head, shown partially open.

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TABLE I.—*Size, weight, and capacity requirements*

Size	Overall length (Min.)	Handle designs (optional)	Weight (Min.)	Capacity (at least)
No.	Inches	No.	pounds	Inches
20	20	1	2	1
20	20	2	2	1
20	20	3	1-1/2	1
20	20	4	1-1/2	1
24	24	1	2-1/2	1-1/4
24	24	2	2-1/2	1-1/4
24	24	3	2	1-1/4
24	24	4	2	1-1/4
27	27	1	2-3/4	1-1/2
27	27	2	2-3/4	1-1/2
30	30	1	3	1-3/4
30	30	2	3	1-3/4

3.8 **Size, weight, and capacity requirements.** The sizes of pruning shears covered by this specification shall be determined by the minimum overall length, measured as indicated in figure 1. The minimum weight of each shear, as applicable to the handle design, shall be in accordance with table I. The capacity of the shear shall be determined by cutting live tree branches of the minimum capacity size specified in table I, without taking more than one "bite" with the shears. The capacity of each shear shall be determined with the two handles opened to an included angle not greater than 90 degrees. The hook on style A cutting heads shall cradle the applicable capacity size branches. The vee opening in style B opened cutting heads shall contain the applicable capacity size branches.

3.9 **Repair parts.** The successful bidders supplying pruning shears covered by this specification shall make available and furnish repair parts when requested. All parts having the same manufacturer's part number shall be constructed to definite standards, tolerances and clearances, in order that such parts may be interchangeable, replaced, or adjusted without requiring modification.

3.10 **Workmanship.** Workmanship shall be first class in all respects. Shears shall be smoothly and properly finished, and free from any defects which may adversely affect their durability and serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Inspection responsibility.** The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Sampling for lot acceptance.**

4.2.1 *Inspection lot.* All shears of one size presented at one time shall be considered a lot for purposes of acceptance inspection and tests.

4.2.2 *Sampling for inspection.* A random sample of shears shall be taken from each inspection lot of material offered for inspection of visual and dimensional characteristics with lot acceptance based on the sampling inspection requirements of table II in accordance with the procedures of MIL-STD-105.

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TABLE II.—*Sampling for visual and dimensional inspection of shears AQL (approx.) =2.5 percent defective*

Number of shears in inspection lot	Sample size, number of shears to be inspected (reference Military Standard MIL-STD-105)
40 and under ..	Inspection level III
41 to 300	Inspection level II
301 and over	Inspection level I

4.2.2.1 *Sampling for tests.* A random sample of shears shall be taken from each inspection lot in accordance with table III, and shall be subjected to the cutting, hardness, and moisture tests specified in 4.4.1, 4.4.2, and 4.4.3, respectively.

TABLE III.—*Sampling for tests*

Number of shears in inspection lot	Number of shears in sample	Acceptance number (failures)	Rejection number (failures)
15 and under .	3	0	1
16 to 40	5	0	1
41 to 110	7	0	1
111 to 300	10	0	1
301 to 800	15	1	2
801 to 1,300 ..	20	1	2
1,301 and over ..	30	2	3

4.3 Inspection.

4.3.1 *Visual and dimensional inspection.* Each of the sample shears selected in accordance with table II shall be visually and dimensionally inspected by the inspector to verify compliance with this specification. Any shear in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defectives in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Rejected lots may be offered again for inspection provided the contractor has repaired or removed all nonconforming shears. The inspector shall again select and examine samples from such resubmitted lots to verify compliance with this specification.

4.3.2 *Testing.* Sample shears selected in accordance with table III shall be subjected to the tests specified in 4.4. Any shear failing any test shall be rejected, and if the number of defectives in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.3.3 *Examination of the end item.* Examination of the pruning shears shall be in accordance with the classification of defects specified in table IV. The lot size shall be expressed in units of pruning shears for the purpose of determining the sample size in accordance with MIL-STD-105. The sample unit shall be one completely fabricated pruning shears. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in defects per hundred units.

TABLE IV.—*Classification of defects*

Examine	Defect	Major	Minor
Finish	Peeling or blistering		X
	Not finished as specified		X
	Area of thin film or no film		X
	Sags, runs, orange peel, wrinkles		X
	Dirt, grit or foreign matter embedded in finish		X
	Finish tacky (not completely dry) ..		X
	Touch up not neat affecting appearance		X
	Surfaces not free from pits, cracks, or hard spots ...		X
	Surface not properly cleaned before painting ...		X
	Castings or forgings not free from blow-holes, porosity, hard spots, shrinkage defects, cracks, or other injurious defects ..	X	

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Examine	Defect	Major	Minor
Finish (cont'd)	Surface not smoothly finished where required		X
Materials	Not fabricated of the specified material	X	
Design	Characteristic not in accordance with the specified requirements	X	
Components	Fractured, split, punctured, sprung, malformed, or otherwise impaired	X	
Workmanship (general)	Sharp burr or sliver on surface to be handled during operation	X	
	Operation omitted or not properly performed, affecting serviceability	X	
	Functioning component that is inoperative or will not function as intended	X	
	Parts not interchangeable	X	
	Operation omitted or not properly performed, not affecting serviceability .		X
	Not connected or joined as specified or poorly accomplished		X
	Assembly is not properly adjusted to perform the function intended		X
Bolts, nuts, screws and rivets	Missing, broken, stripped, or fractured	X	
Assembly	Component not properly assembled seriously affecting serviceability, e.g., threaded component cross threaded	X	
Marking	Missing, illegible, or otherwise incorrect		X

4.4 Test procedures.

4.4.1 *Cutting test.* Shears shall be tested for cutting ability by severing at least 15 branches of not less than the specified capacity (branch diameter) of green wood such as ash, beech, hickory, or pine. After being subjected to this test, the shears shall show no sign of distortion to any part, damage to the cutting edge, or indication of failure in any respect. The shears shall be well balanced, operate smoothly, and sever branches with a minimum of effort. Shears failing these tests shall be rejected.

4.4.2 *Hardness test.* Cutting blades of shears selected for the cutting test shall be tested for compliance with the hardness requirements specified in 3.1.2. Any cutting blade which does not meet the respective hardness requirement shall cause the shears to be rejected.

4.4.3 *Moisture content.* Determination of moisture content of wood handles and sections (see 3.1.1) shall be made by a moisture meter or other standard method.

5. PREPARATION FOR DELIVERY

(Civil agencies. Fed. Std. No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.)

5.1 Preservation and packaging.

5.1.1 *Level A.* Level A cleaning, preservation, and packaging shall be in accordance with level A of MIL-H-15424.

5.1.2 *Level C.* Level C cleaning, preservation, and packaging shall be in accordance with the contractor's commercial practice.

5.2 Packing.

5.2.1 *Level A.* Level A packing shall be in accordance with level A of MIL-H-15424.

5.2.2 *Level B.* Level B packing shall be in accordance with level B of MIL-H-15424.

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5.2.3 *Level C.* Pruning shears, cleaned, preserved, and packaged as specified in 5.1 shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with the rules or regulations of carriers as applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking specified in the contract or order (see 6.1), each unit and intermediate package and shipping container shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to any special marking specified in the contract or order (see 6.1), interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Size of shears required (see 1.2.1).
(*c*) The various handle designs and cutting heads are generally satisfactory for the purpose intended, however, if the procuring agency can justify specific handle designs or cutting head styles, the applicable options should be exercised (see 1.2.2, 1.2.3, and 3.5).
(*d*) If either a bolted or riveted cutting head is required, so state (see 3.1.2.1).
(*e*) State the levels of packaging and packing required (see 5.1 and 5.2).
(*f*) Special marking, if required (see 5.3).

6.1.1 When ordering repair parts (see 3.9), the following should be specified:

(*a*) The manufacturer's name or trademark of the shears for which the parts are required.
(*b*) Size of shears for which the part is required.
(*c*) Name and description, if necessary, of the part required.
(*d*) Quantity of each item required.

6.2 Transportation description. Transportation description and minimum weights applicable to this commodity are:

Rail:

Pruners, tree.
Carload minimum weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Pruners, tree.
Truckload minimum weight 24,000 pounds, subject to Rule 115, National Motor Freight Classification.

6.3 Superseding data. This specification supersedes type IV hand pruning shears covered by GGG-T-669a.

DOD coordination has been waived.

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